Cartpole Problem Analysis

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CS-370: Current and Emerging Trends in Computer Science

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Explain how reinforcement learning concepts apply to the cartpole problem.

Reinforcement learning is a type of machine learning where an intelligent agent learns how to make choices that maximize reward through trial and error. Specifically, this project used the Deep Q-Learning algorithm. In this project, the goal for the agent was to learn how to solve the cartpole problem, a task where the agent has to balance a pole on an unstable cart. The possible state values are cart position, cart velocity, pole angle, and pole tip velocity. The action values are moving to the left or moving to the right.

Analyze how experience replay is applied to the cartpole problem.

Experience replay refers to an agent storing previous attempts to complete a task in memory and randomly retrieving these attempts to use as samples that they can train with. In this case, the agent is "remembering" past experiences with the cartpole problem to train their neural network into becoming better at solving the problem. Additionally, discount factors are used to set the agent's preferences towards future rewards rather than immediate rewards.

Analyze how neural networks are used in deep Q-learning.

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This project uses a neural network architecture where the network predicts and updates the Q-values used by the agent continuously. This technique makes the algorithm more efficient and scalable. When I slightly increased the Hyperparameter values in the second program, the algorithm went through at least twice the number of runs before reaching the state as the program did when left as-is.

References

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